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Hydraulic Property Estimation of Rock Fractures Using Deep Learning

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Abstract

Accurate estimation of hydraulic aperture and permeability in fractured rocks is essential for modeling fluid flow in subsurface systems but remains challenging due to the geometric complexity of fracture surfaces. Traditional models, including the classical cubic law and its variants, often rely on simplified assumptions and fail to account for roughness, tortuosity, and contact areas, resulting in significant prediction errors. While high-fidelity numerical simulations based on the Navier–Stokes equations provide more accurate estimates, they are computationally expensive and impractical for large-scale applications. In this study, we present a convolutional neural network (CNN)-based image-to-value regression model that predicts the hydraulic aperture of three-dimensional fractures from mechanical aperture maps. A diverse dataset of 1000 synthetic fractures was generated using PySimFrac. True hydraulic apertures were computed using computational fluid dynamics (CFD) simulations in COMSOL Multiphysics and served as ground truth labels for supervised learning. The CNN was trained on the image-label pairs and evaluated using both validation and blind test datasets. The model achieved high predictive accuracy with R^2 values exceeding 0.99 during validation and blind testing. Furthermore, on a new dataset, the trained model achieved high prediction accuracy with a mean relative error of 0.4%, outperforming four benchmark cubic-law-based models. Additionally, the CNN exhibited remarkable stability across a wide range of fracture roughness, unlike traditional models whose errors increased significantly with increasing roughness. This study demonstrates the feasibility and effectiveness of deep learning models as a fast, scalable, and accurate alternative to traditional flow-based and empirical upscaling methods for fracture hydraulic properties.

Introduction

Fluid flow through fractured geological media plays a critical role in numerous subsurface applications, including groundwater management, hydrocarbon extraction, geothermal energy production, and carbon dioxide sequestration (Cao et al., 2022; Gong et al., 2023; March et al., 2018; Streltsova, 1976; Warren & Root, 1963; P. Witherspoon, 1986). Accurately characterizing the hydraulic properties of fractures, particularly hydraulic aperture and effective permeability, is essential for modeling and predicting flow behavior in fractured rock systems (Bear et al., 2012; Berre et al., 2019; Dietrich et al., 2005). However, due

to the inherent geometric complexity of natural fractures, direct measurement of these properties is often impractical or infeasible, especially at larger scales.

Traditionally, analytical or semi-empirical models have been used when modeling fluid flow in fractures. The most commonly used model is the cubic law, which relates the volumetric flow rate through a fracture to the cube of its mechanical aperture. The cubic law idealizes fractures as two smooth parallel plates with a uniform aperture (Snow, 1969). Several studies have shown that the assumptions underlying the cubic law often lead to significant overestimation of permeability, especially in rough-walled fractures or those with contact areas, high tortuosity, and varying aperture distributions (Brown, 1987; He, Sinan, et al., 2021; Tsang, 1984; L. Wang et al., 2015; Witherspoon et al., 1980; R. W. Zimmerman & Bodvarsson, 1996).

As a result, several corrections have been proposed that attempt to account for complex fracture geometry by: 1) modifying the classical cubic law, or 2) implementing a local cubic law model. The first group approaches the problem by either modifying the definition of the aperture used in the cubic law to either an arithmetic mean, geometric mean, or harmonic mean aperture (Brown, 1987; Renshaw, 1995; Unger & Mase, 1993), or by incorporating additional correction factors to account for fracture roughness and tortuosity (Patir & Cheng, 1978; Renshaw, 1995; Z. Wang et al., 2018; R. W. Zimmerman & Bodvarsson, 1996). The second group implements a local cubic law that divides the fracture into a grid of local elements, each with its own locally defined aperture value. The cubic law is then applied at each element, and the overall flow is obtained by solving the pressure field across the domain using mass conservation principles (Brush & Thomson, 2003; He, Sinan, et al., 2021; Oron & Berkowitz, 1998; L. Wang et al., 2015). However, these corrections and adaptations to the cubic law remain limited in their applicability. The global modification of the classic cubic law shows inaccuracies when dealing with complex fracture geometries. The local cubic law models offer a better approximation of flow behavior but are more computationally expensive because of the need to generate a mesh and solve a flow problem.

For fractures with complex geometries, high-fidelity numerical solutions of the Navier-Stokes equations provide physically accurate flow predictions (Crandall et al., 2010; Koyama et al., 2008; Xie et al., 2015). These methods, typically implemented via computational fluid dynamics (CFD) tools, resolve the fluid velocity field throughout the fracture domain and account for inertial effects, eddies, and localized accelerations. However, CFD simulations are computationally expensive and time-consuming, thus limiting their application in large-scale or real-time workflows (Brush & Thomson, 2003; L. Wang et al., 2015; R. Zimmerman & Yeo, 2000). While high-resolution Navier-Stokes simulations overcome the inaccuracies of cubic law models, they remain computationally inefficient. Consequently, it has become crucial to develop a model to efficiently and accurately upscale fracture hydraulic properties.

Recent advances in machine learning techniques have inspired numerous applications in geosciences, including the prediction of fracture hydraulic properties. Several studies have shown that with good quality training data and a well-designed network architecture, data-driven models outperform cubic-law-based models in accuracy while presenting computational efficiency superior to flow-based upscaling methods (He et al., 2021; Ma et al., 2025; Zhu et al., 2023). In particular, convolutional neural networks (CNNs) are capable of recognizing patterns in images and extracting complex spatial features, potentially providing a powerful tool in predicting fracture hydraulic properties from fracture images. Notably, with the advancement of high-resolution imaging techniques capable of capturing complex fracture geometries, there are limited tools to reconcile the high-resolution fracture images and the hydraulic properties of the fractures within the upscaling workflows. We therefore propose a CNN-based framework for predicting fracture hydraulic properties from images of three-dimensional fractures.

In this study, we develop a CNN-based image-to-value model to map the complex non-linear relationship between high-fidelity images of rock fractures as input, and fracture hydraulic aperture as output. This approach provides the capability to process image-based input of rock fractures and map them to numerical labeling reflecting the hydraulic properties of rock fractures. Our proposed workflow to develop the model is divided into three main steps. First, we generate an extensive dataset of complex fractures and perform high-

fidelity Navier-Stokes simulations with parallel computing to predict the flow-based hydraulic apertures of the fractures. Next, we train and optimize a CNN using the generated dataset of fracture images and hydraulic apertures. Finally, we perform blind testing of the optimized CNN model on a new dataset and benchmark its performance against reference Navier-Stokes solutions. After testing and validation, the trained model offers a robust alternative to traditional upscaling methods that can be readily integrated into reservoir characterization workflows.

Dataset Generation

Fracture generation

To train a robust convolutional neural network (CNN) capable of predicting the hydraulic aperture and effective permeability of fractures from their geometrical features, we developed a high-quality synthetic fracture dataset that combines statistical fracture generation with physics-based flow simulation. To identify the parameters for fracture generation, we employed Latin Hypercube Sampling (LHS) to efficiently explore a broad, multidimensional parameter space relevant to fracture morphology. The considered fracture parameters are summarized in [Table 1](#).

Table 1—Parameters used in LHS to generate fractures in PySimFrac.

Parameter	Range	Distribution
Roughness*	[0.1 – 0.9]	Normal
Mean Mechanical Aperture	[0.1 – 5]	Log-normal
Hurst Exponent	[0 – 1]	Normal
Anisotropy	[0 – 1]	Normal
Mismatch Length Scale (Fraction of Fracture Size)	[0.1 – 1]	Normal

* Roughness is described as the standard deviation of the fracture surface heights from a mean plane.

Each parameter set generated through LHS was used as input to PySimFrac, a Python-based stochastic fracture generation toolkit for creating complex synthetic fractures using either a Fourier-based spectral method or a convolution-based method ([Guiltinger et al., 2024](#)). We used the spectral method, which creates self-affine fracture surfaces with prescribed statistical properties by manipulating the power spectral density in the frequency domain. The physical dimensions of the generated fracture surfaces were 50×50 mm, with a discretization of 1×1 mm. We generated 1000 three-dimensional synthetic fractures with diverse and realistic geometrical features. Each synthetic fracture consists of a pair of 3D surfaces representing the opposing walls of each fracture, from which the mechanical aperture field was computed. A sample synthetic fracture generated using PySimFrac is shown in [Figure 2](#).

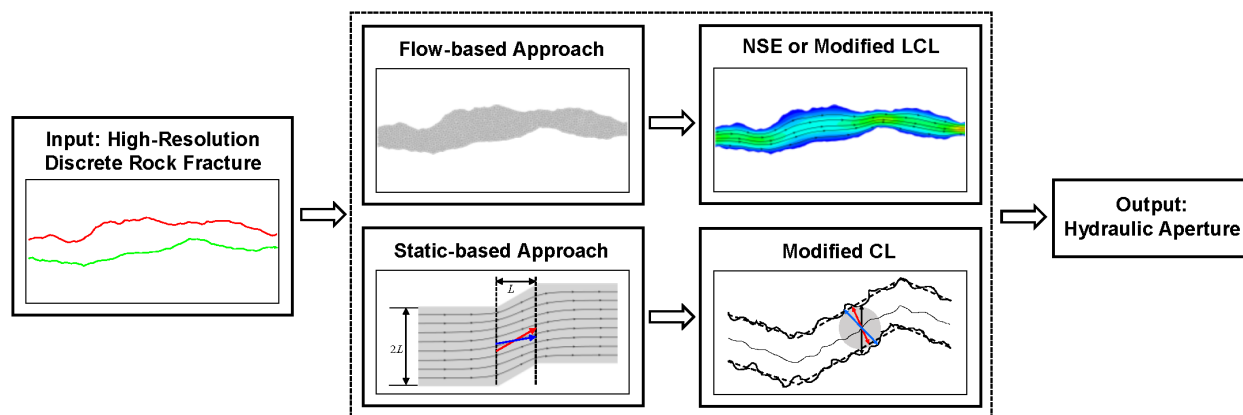


Figure 1—Traditional fracture upscaling workflow, highlighting flow-based methods (top) and static-based methods (bottom), after He (2022).

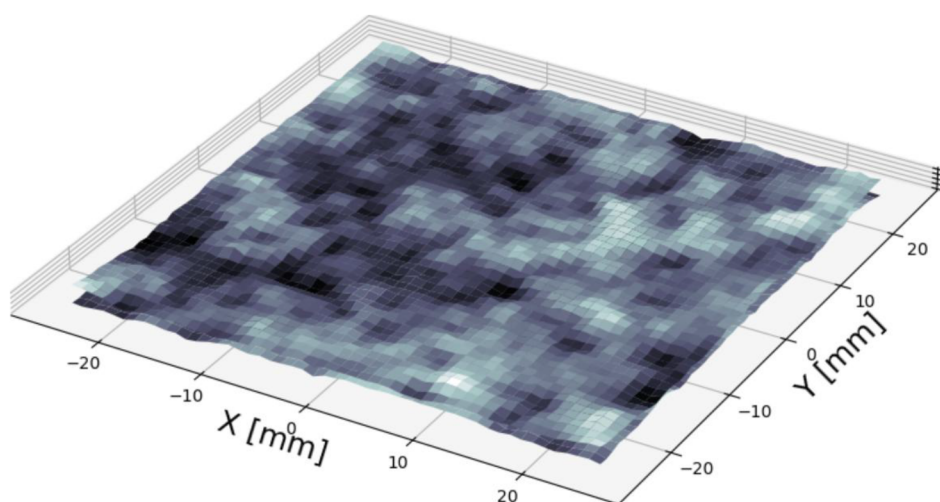


Figure 2—Sample fracture generated using PySimFrac.

Computational fluid dynamics (CFD) simulations

Each synthetic fracture was imported into COMSOL Multiphysics v5.4, and a high-resolution tetrahedral mesh was generated to capture the fracture geometry, as shown in Figure 3. The Navier-Stokes equations were numerically solved for steady-state laminar flow of a single-phase, incompressible fluid through the fracture. The fluid had a dynamic viscosity of 0.001 Pa.s (1 cP) and a density of 1000 kg/m³. The initial fluid velocity was 0 m/s in all directions, and the initial domain pressure was set to 0 Pa. One side of the fracture was designated as a flow inlet at a fixed pressure of 0.1 Pa, and the opposite end as an outlet at 0 Pa, creating a pressure gradient across the fracture. All other fracture surfaces were set to no-flow and no-slip boundary conditions. The steady-state pressure and velocity solutions were obtained by solving the Navier-Stokes equations. The flux was obtained by integrating the velocity solution over the outlet cross-section. The obtained flux was used to calculate the equivalent hydraulic aperture of the fracture and effective permeability following the equation below:

$$k_{eff} = \frac{a_h^2}{12} = (12)^{\frac{1}{3}} \times \left(\frac{Q\mu}{w\nabla p} \right)^{\frac{2}{3}}$$

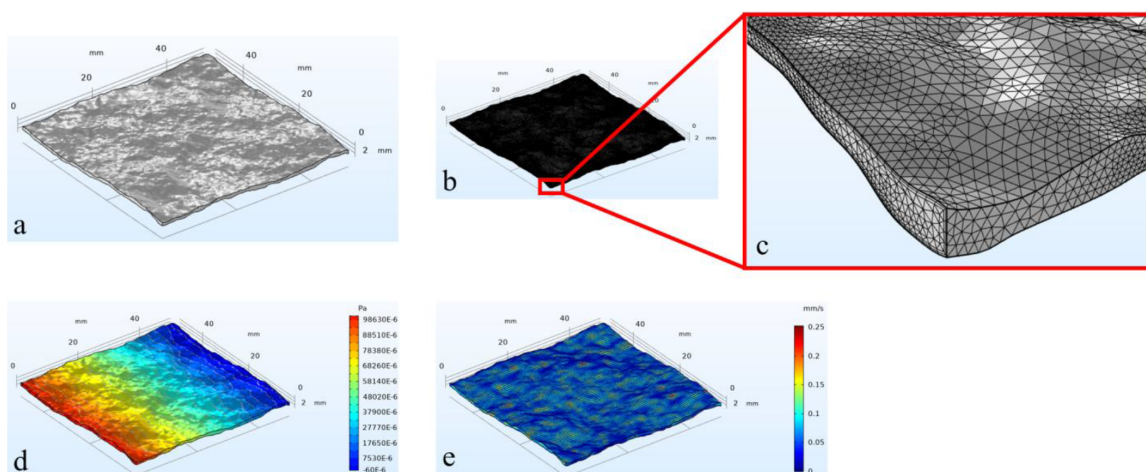


Figure 3—COMSOL workflow to obtain the flow-based estimation of the fracture hydraulic aperture. a) imported digital fracture, b) & c) high resolution meshing, d) steady-state pressure solution, e) velocity solution

where k_{eff} is the effective permeability, a_h is the hydraulic aperture, Q is the volumetric flux, μ is the fluid viscosity, w is the fracture width, and ∇p is the pressure gradient across the fracture.

For each fracture, 2D maps of the mechanical aperture were extracted and saved as images. Each image was paired with a corresponding hydraulic aperture value calculated from the COMSOL simulations, forming an image-label pair. This dataset was used to train and validate the developed CNN.

Convolutional Neural Network (CNN) Development

Image pre-processing

The 2D fracture images underwent several preprocessing steps before being used to train the CNN. First, each fracture image was converted to a 32×32 -pixel image for training efficiency. Next, each image was converted to grayscale with the mechanical aperture represented as grayscale intensity values between 0 and 255, as shown in Figure 4. The grayscale intensities were normalized by the maximum mechanical aperture value of the dataset, such that this maximum value corresponded to a grayscale intensity of 255. Next, the grayscale images were converted to PyTorch tensors and the pixel values were scaled to within the range -1 to 1.

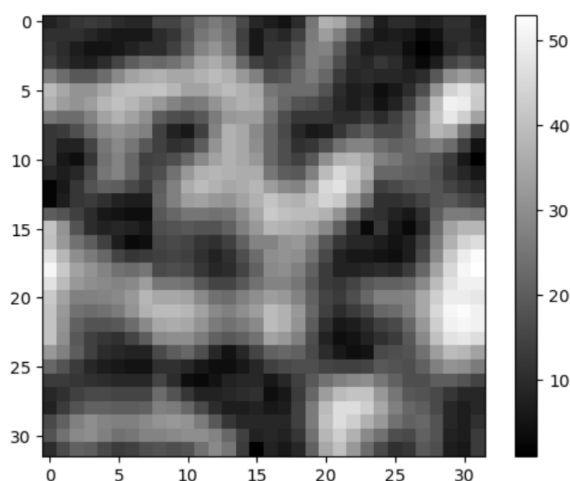


Figure 4—A sample fracture image with a 32×32 -pixel resolution converted to grayscale with the mechanical aperture represented as grayscale intensity between 0 and 255.

CNN architecture and training

We used PyTorch (Paszke et al., 2019) to build and train a CNN capable of learning the mapping from mechanical aperture images to scalar outputs representing hydraulic properties. The CNN architecture was designed to extract hierarchical spatial features from the input image and regress to a single output value. The dataset of 1000 diverse fractures was used in the training of the neural network. The dataset was split into training data (70%), validation data (20%), and blind testing data (10%).

The CNN architecture, illustrated in Figure 5, comprises an input layer that accepts a single-channel 32 × 32-pixel image, followed by three convolution blocks, and two fully connected layers. Each convolution block includes a 2D convolution function for feature extraction, a ReLU activation function that introduces non-linearity so the network can model complex relationships, and a Max-Pooling function for spatial down-sampling. Through these layers, the network will have extracted hierarchical features, progressively reducing the spatial size but increasing the feature richness. The resulting feature maps are flattened and passed through two fully connected layers: the first with 128 neurons and ReLU activation, and the second with a single linear output neuron for regression. The final output of the CNN is a prediction of the hydraulic aperture of the fracture based on the image input.

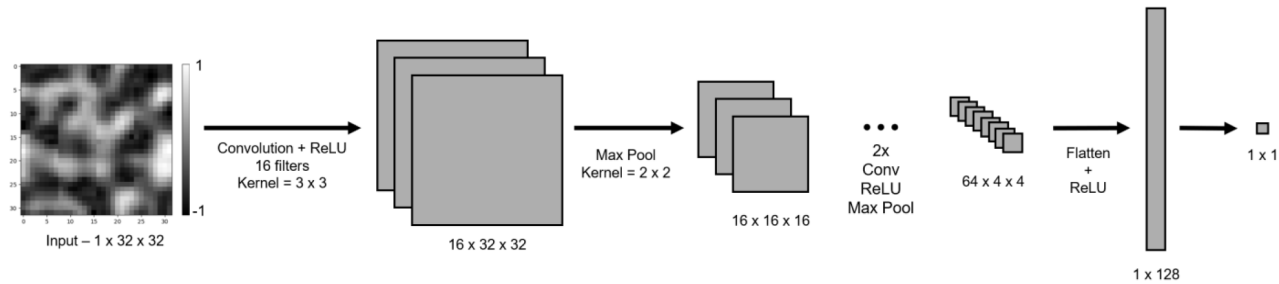


Figure 5—The CNN architecture showing the input layer, convolution blocks, and fully connected layers.

The CNN was trained using the adaptive moment estimation (Adam) algorithm for optimization. The optimizer updates the weights and biases of each layer during the training process to improve the prediction accuracy using a backpropagation algorithm. The initial learning rate was set to 0.001, and the model performance was evaluated using Mean Squared Error (MSE) loss, which is ideal for regression problems. The maximum epochs allowed in training is 100. However, early stopping is implemented to prevent overfitting by stopping training when the validation loss ceases to improve, ensuring optimal model generalization. The training hyperparameters are summarized in Table 2.

Table 2—CNN training hyperparameters.

Hyperparameter	Value
Optimizer	Adam
Initial Learning Rate	0.001
Batch Size	32
Kernel Size (Convolution/Max Pool)	3/2
Kernel Stride (Convolution/Max Pool)	1/2
Padding	1
Early Stop Patience	10 epochs
Maximum Epochs	100

Results and Discussion

CNN training and validation

We developed a deep learning-based approach that leverages convolutional neural networks (CNNs) to predict the hydraulic properties of fractures with complex geometries from images of their mechanical aperture. 1000 image-label pairs were used to train, validate, and test a robust CNN for predicting hydraulic aperture. Figure 6 shows the training and validation loss curves, in terms of mean squared error (MSE), over 45 epochs. Both curves show a sharp decline during the initial epochs, indicating rapid convergence of the model. Notably, there is no divergence between training and validation losses, suggesting that the model generalizes well and is not overfitting.

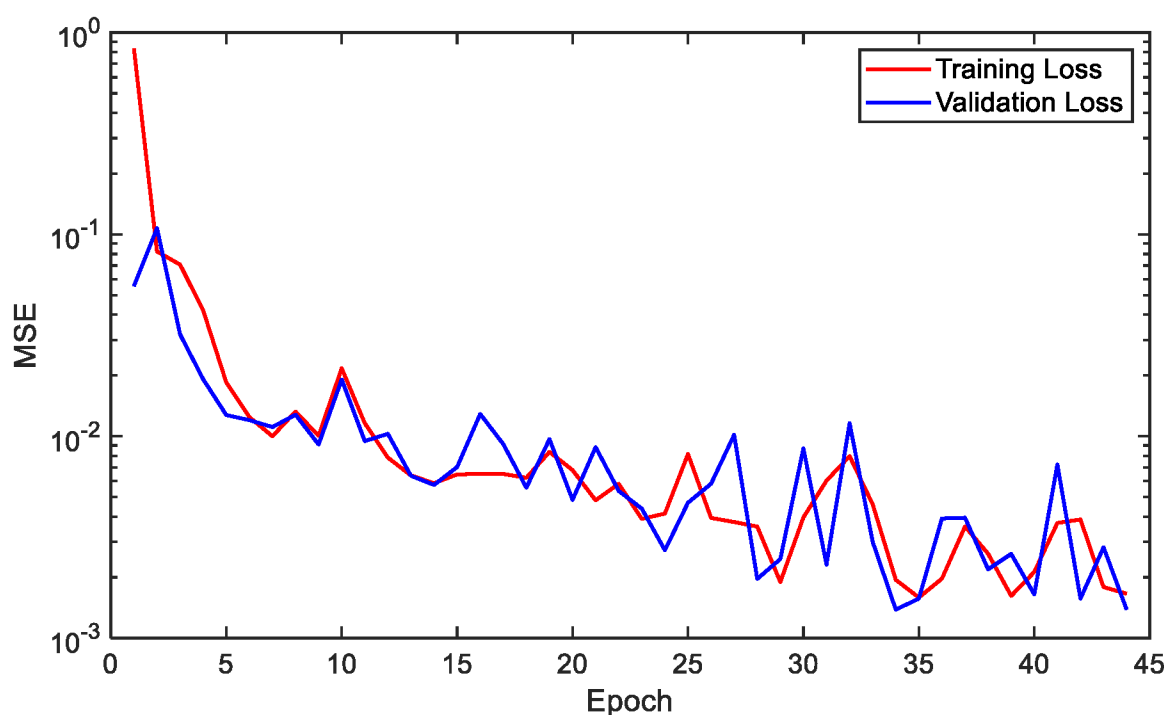


Figure 6—Training and validation performance of the CNN.

The trained model's ability to accurately predict hydraulic aperture was compared against numerical solutions of the Navier-Stokes Equations, considered as the true hydraulic aperture in this study. This comparison was made for the validation dataset and an unseen dataset during training or validation, as presented in Figure 7. The trained model's predictions are highly accurate with R^2 values greater than 0.99 for both validation and blind testing data sets. The trained CNN's robust performance in blind testing highlights the model's capability to generalize beyond the training dataset, a key feature important for practical deployment in reservoir characterization workflows.

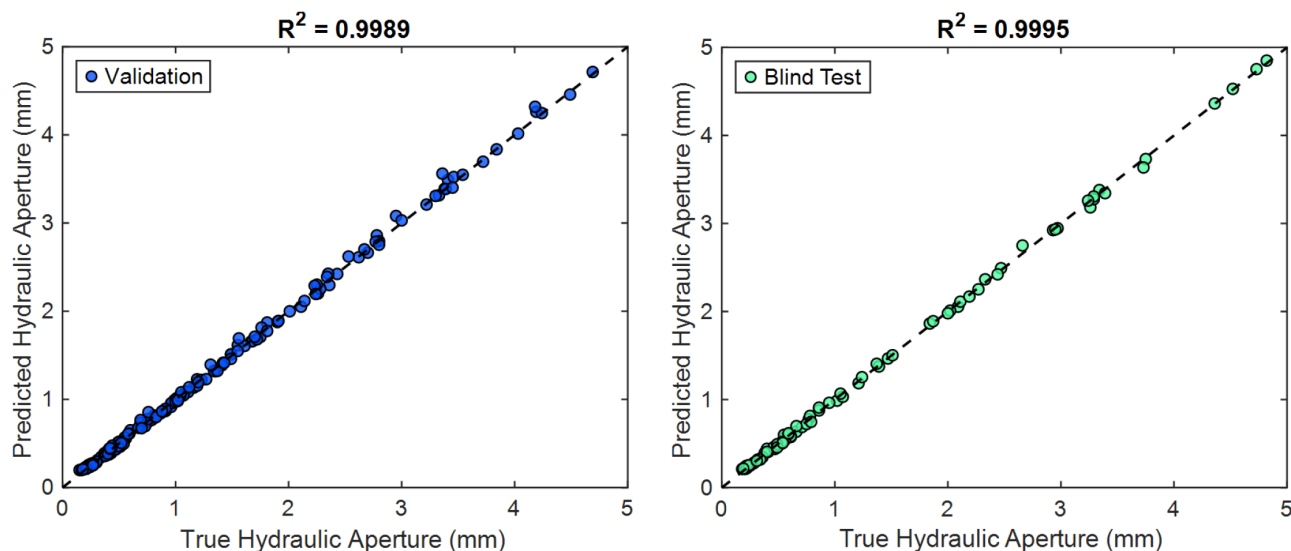


Figure 7—Comparison of CNN predictions against CFD simulations for validation (left) and blind test datasets (right).

CNN benchmarking

We compared our trained model against four popular cubic-law-based models from literature using a new dataset of 100 diverse and complex fractures with roughness ranging between 0.1 and 0.9 mm. The cubic-law-based models implemented in this comparison are listed in Table 3.

Table 3—Cubic-law-based models from literature used in benchmarking comparison.

Reference	Model	Description
Brown (1987)	$a_h = \langle a_m \rangle$	Arithmetic mean of normal apertures
Konzuk & Kueper (2004)	$a_h = \text{geomean}(a_m)$	Derived based on perturbation analysis; Geometric mean of normal apertures
Renshaw (1995)	$a_h = \langle a_m \rangle \left[1 + \left(\frac{\sigma_{a_m}}{\langle a_m \rangle} \right)^2 \right]^{-0.5}$	Derived based on lognormally distributed fracture apertures
Patir & Cheng (1978)	$a_h = \langle a_m \rangle \left[1 - 0.9e^{-\frac{0.56 \langle a_m \rangle}{\sigma_{a_m}}} \right]^{\frac{1}{3}}$	Empirical model fitted with numerical results from solving Reynolds equation

Figure 8 shows the distributions of relative error for all tested models. The relative errors are calculated compared to the Navier-Stokes solution following the equation:

$$\text{Relative Error (\%)} = \frac{a_{hi} - a_{NSE}}{a_{NSE}} \times 100$$

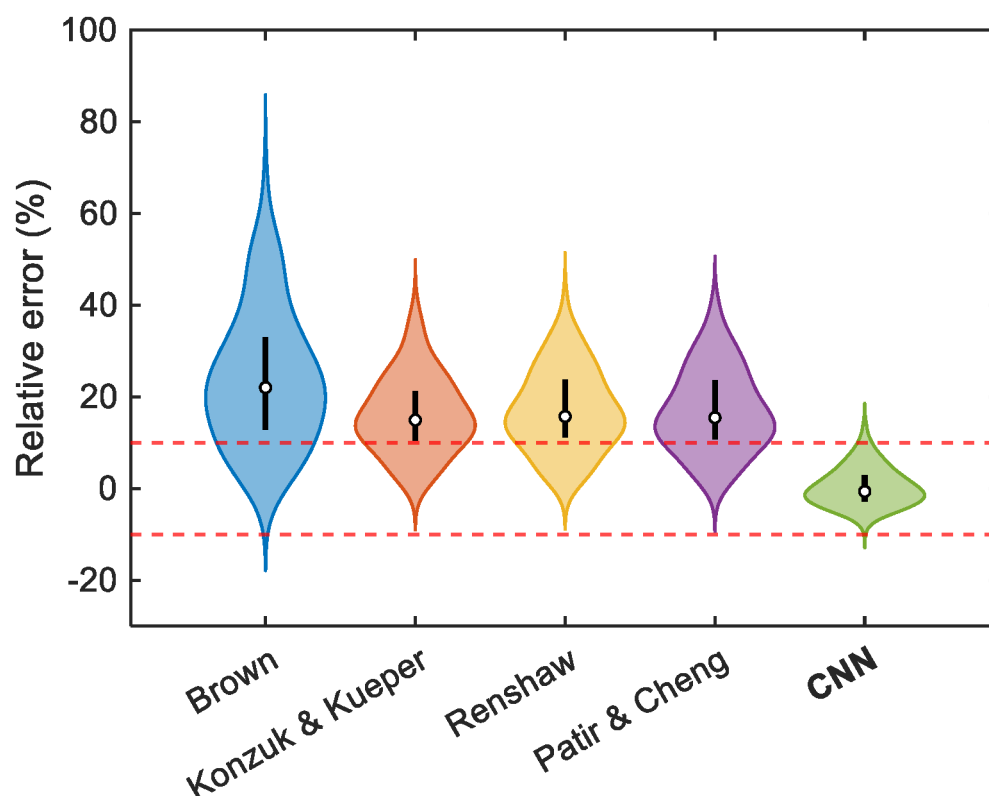


Figure 8—Relative error between Navier-Stokes hydraulic aperture (true hydraulic aperture) and different models. The central white-circle mark represents the median (50%), while the black-line bottom and top edges are the 25th and 75th percentiles, respectively. The red dotted lines indicate $\pm 10\%$ relative error.

where a_{hi} is the hydraulic aperture predicted by model i , and a_{NSE} is the hydraulic aperture calculated from the Navier-Stokes simulations. The relative error is calculated such that a positive value indicates an overestimation of the hydraulic aperture while a negative value indicates an underestimation of the hydraulic aperture.

Results indicate that all cubic-law-based models in this comparison tend to overestimate the hydraulic aperture of the fractures. These models show a one-sided distribution with most points above the zero-error axis. In contrast, the CNN model in this study shows a relative error distribution majorly within $\pm 10\%$. The CNN model also shows a normal error distribution centered around the zero point, in contrast to the cubic-law-based models, indicating a much better prediction accuracy. The CNN had a mean relative error of 0.4%, and a standard deviation of 4.2%. In comparison, the mean relative errors of the cubic-law-based models were 23.7%, 16.1%, 17.0%, and 17.0% for the Brown, Konzuk & Kueper, Renshaw, and Patir & Cheng models, respectively. Their respective standard deviations were 14.5%, 8.3%, 8.9%, and 8.9%.

We further compared the stability of the trained CNN against the cubic-law-based models by considering the change of absolute relative error with varying fracture roughness, as shown in Figure 9. Results show that the trained CNN is stable over the investigated roughness range, showing a low absolute relative error and minimal variation with changing roughness. In contrast, the cubic-law-based models all show higher absolute relative errors than our CNN. In addition, their absolute relative errors increase significantly with increasing roughness, indicating poor performance for prediction of hydraulic aperture in fractures with high roughness.

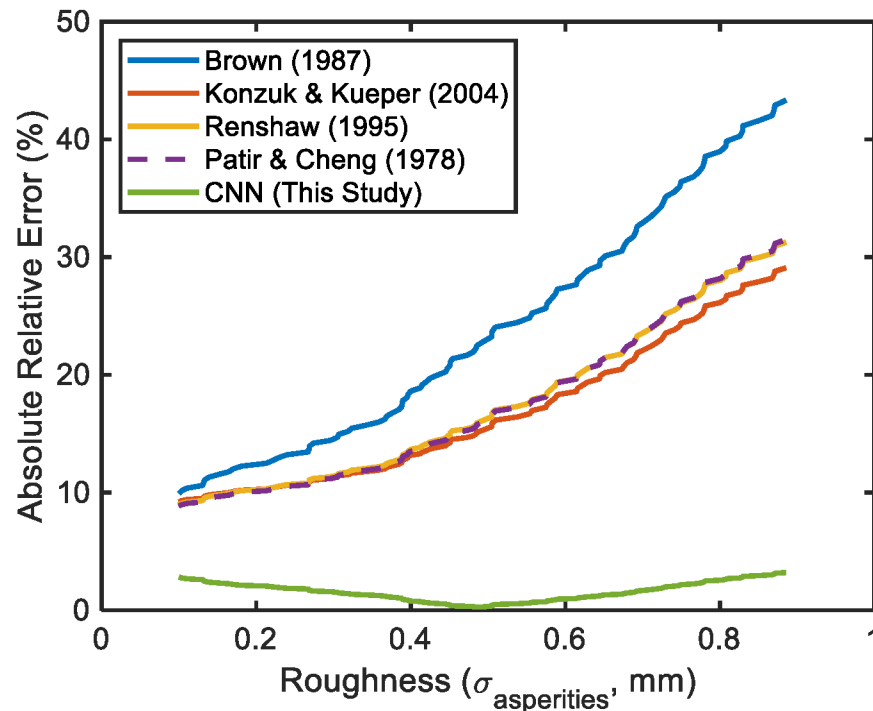


Figure 9—Absolute relative error as a function of fracture roughness for the cubic-law-based models, and the trained CNN. The fracture roughness is quantified by the standard deviation of surface heights from a mean plane.

Conclusions

In this work, we develop an image-based convolutional neural network (CNN) model for predicting the hydraulic aperture of rock fractures. The CNN was trained on a large, synthetically generated dataset of complex 3D fracture geometries, whose true hydraulic apertures were calculated from high-fidelity Navier–Stokes simulations. The results demonstrate that:

- The trained CNN achieves high prediction accuracy, with R^2 values exceeding 0.99 on both validation and blind testing datasets.
- Compared to four traditional cubic-law-based models, the CNN showed significantly improved performance. The cubic-law-based models consistently overestimated hydraulic aperture and exhibited higher mean relative errors (16–24%) and broader variability. In contrast, the CNN maintained a lower mean relative error (0.4%) and less variability.
- The trained CNN showed robustness across a wide range of fracture roughness, maintaining stable predictive performance even for highly rough fractures. Traditional cubic-law-based models, in comparison, showed increasing error with increasing roughness, suggesting limited application for high roughness fractures.

The proposed image-based approach offers an accurate and computationally efficient surrogate for flow simulation. Once trained, the CNN can deliver accurate and computationally efficient predictions of hydraulic aperture from fracture images, outperforming traditional cubic-law-based models and eliminating the need for high-fidelity CFD simulations in reservoir characterization workflows. Overall, this study highlights the potential of deep learning to transform fracture characterization by bridging the gap between high-resolution imaging and accurate flow property estimation.

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